Research Project: Results and Recommendations

Renesas Capstone

Root Cause Analysis Model for Semiconductor Mass Production Failures

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Results

The purpose of our capstone project and research for Renesas was to enhance their existing Root Cause Analysis (RCA) process by developing a multivariate, model-driven framework capable of identifying the electrical tests most predictive of Category C wafer failures. The overall objective of our research is to describe failure test patterns and generate a ranking of the most predictive tests that can guide Renesas' team towards better engineering of their methods for process improvement, test optimization, and more reliable production. The company became interested in diagnosing the rise of Category C test failures that began in July 2024. Wafer Process Data (WAT) and Wafer Sort Test Data (WT) together generate hundreds of electrical measurements per die and the foundation of RCA.

Our objective was to design a more robust and scalable RCA approach that could surface feature-level reasoning and generate deeper insight into which tests drive yield loss. To achieve this, we built a feature-selection pipeline using three complementary machine-learning models: Random Forest, XGBoost, and Lasso Regression. We created an aggregated, ensemble-based importance score for every electrical test by averaging the ranked feature importance scores across the three models. This aggregation reduces model-specific bias, avoids overfitting to any single algorithm's tendencies, and produces a more stable, interpretable ranking of influential tests for Renesas. From this unified ranking, we identified the top 50 most important electrical tests as the most impactful subset to build on as the foundation for a Decision Tree–based multivariate RCA model. The decision tree provides clear, rule-based pathways illustrating how combinations of electrical tests and threshold values separate passing dies from failing ones. This allowed us to extract cutoff scores, feature interactions, and branching logic to clearly interpret why failures occur and which specific patterns of test values produce risk at the die level.

The exploratory analysis and multivariate modeling reveals several systematic mechanisms driving Category C yield loss as outlined below.

Category C Failure Distribution and Magnitude

Overall failure distribution analysis (Figure 1) reveals approximately 15% of the tests were failures, and Category C failures dominate the yield loss profile, accounting for 73% of all probe stage failures. Because high failure is observed across multiple months of production, Category C high failure represents a persistent, systemic yield limiter rather than an intermittent issue, similar to dominant failure classes reported in prior yield studies (Hsu et al.; Riordan et al.).

Exploratory Data Analysis: Spatial, Temporal, and Equipment Patterns

Wafer Position Effects

Yield as a function of wafer position within lots (Figure 2) reveals significant first wafer instability. Wafer Number 1 exhibits a failure rate of roughly 28%, substantially higher than subsequent wafers, which show lower and more stable rates. This pattern indicates that process conditions at the beginning of lot processing have not yet reached a steady-state. It suggests a systematic opportunity for yield improvement via introducing dummy wafers prior to the first production.

Spatial Defect Patterns

Wafer map analysis (Figure 3) shows nonuniform failure distributions with a pronounced edge ring pattern. Zone 4, the outer edge region of wafers, has the highest defect concentration, with failing dies clustering near the periphery while interior regions predominantly pass. This spatial structure is characteristic of lithography focus errors at the field edge, resist coating non-uniformity, or edge-related CMP variation, suggesting the need for targeted edge zone

interventions (Hansen et al.; Stine et al.). Similar edge and ring patterns are widely documented in wafer map classification systems (Jeong et al.; Chien et al.; Wu et al.; Alawieh et al.).

Probe Card Variability

When wafers are grouped by the probe card used during electrical testing (Figure 4), significant variation emerges. Probe Cards 02 and 03 exhibit substantially elevated failure rates. This suggests that part of the measured yield loss is attributable to specific test hardware degradation, such as worn probe tips, contamination, or contact resistance, rather than process defects.

Temporal Patterns and Lot Level Variation

Monthly failure rate trends (Figure 5) show a notable spike in January, during which Category C failures occur at substantially higher rates than in other months. This temporal spike suggests environmental factors, maintenance scheduling, or material lot-to-lot variations that may follow annual cycles, similar to seasonal effects reported in spatio-temporal models (Ahmadi et al.). Across seven fabrication lots (Figure 6), Lots 2 and 4 exhibit elevated failure rates at two to three times the failure rates of other lots. This variation suggests batch-level or tool-specific process instability related to those two lots.

Missing Data Strategy and Procedural Patterns

The analysis reveals 72 skipped tests, primarily in late-stage WT1 sequences, caused by early test failures; once a die fails early in the sequence, subsequent tests are not executed. This creates structured, non-random patterns of missingness. Features with more than 10% missing values are excluded from modeling to maintain data quality and model stability, consistent with best practices for large-scale test data analysis (Daasch and Madge). The modeling pipeline uses

median imputation for missing numeric values combined with missingness indicator flags for tests, preserving the informative nature of skipped tests while avoiding spurious signals.

Multivariate Feature Selection: Model Performance

Three complementary machine-learning models (Lasso, Random Forest, and XGBoost) are optimized using RandomizedSearchCV with ten hyperparameter samples and five-fold stratified cross-validation. Because failures represent only ~15% of the data, all models incorporate a 1:10 class-weighting ratio favoring the failure class, and model performance is evaluated with cross-validation and out-of-sample metrics, including area under the ROC curve (AUC), precision, recall, and weighted F1 score. Given the class imbalance, precision and recall can diverge substantially; therefore, the weighted F1 score is emphasized (Daasch and Madge). This approach follows recent work that applies ensemble learning and feature selection to semiconductor test and wafer data (Kang et al.; Li et al.; Wang et al.; Lenhard et al.).
Lasso logistic regression, using L1 regularization, identifies features with strong linear relationships to Category C failures. Lasso identifies WT1 163541 and WT1 163542 as the strongest linear predictors. It achieves 95% precision and 72% recall for successes and 23% precision with 67% recall for failures, resulting in a weighted F1 of 0.34, serving as the baseline for more flexible nonlinear models.

Random Forest and XGBoost capture nonlinear interactions common in wafer and test data, aligning with recent semiconductor analytics research. Random Forest assigns highest importance to WAT_30_polar, WAT_14_polar, and WAT_24_polar as the strongest predictors (Figure 8), achieving 94% precision with 94% recall for successes and 52% precision with 49% recall for failures, yielding the best weighted F1 score at 0.50. XGBoost similarly validates

WAT_30_polar as the most influential feature across all folds (Figure 9), achieving 92% precision and 97% recall for successes and 61% precision with 36% recall for failures, with an AUC of ~0.77 and weighted F1 of 0.45. Hyperparameter tuning confirms that Random Forest performs best with 200 trees and depth 8, followed by XGBoost (600 trees, depth 4, LR 0.1), while weighted F1 is used as the optimization metric to balance precision–recall tradeoffs under class imbalance.

Cross Model Validation and Feature Overlap

The results of our feature selection showed strong convergence across the three models, especially among WAT and WT1 tests, which consistently appeared at the top of the rankings. This convergence validated the credibility of the ensemble scoring approach and indicated that the highest-ranked tests were truly meaningful contributors to predicting Category C failures. Analysis of the top fifty features from each model (Figure 11) reveals only two features, WAT 42 and WAT 45, appear in the top fifty lists of all three models, marking them as the strongest cross-model signals. Overlap between model pairs is modest: eight features are shared between Random Forest and XGBoost, four between Lasso and Random Forest, and three between Lasso and XGBoost. Each model also contributes a substantial set of unique features, echoing prior findings that different modeling approaches capture complementary aspects of process variation (Reda et al.; Kupp et al.). ROC curve comparison (Figure 12) further validates consistency across methods. Despite differences in feature sets and modeling assumptions, all three models achieve similar AUC values between 0.76 and 0.78, and their ROC curves overlap closely. This suggests that the main predictive signals are robust to the choice of algorithm and reflect genuine process outcome relationships (Kang et al.; Lenhard et al.).

Aggregated Feature Importance Rankings

To obtain a stable feature set for RCA, importance scores from all three models are averaged to produce an aggregated ranking (Figure 13). WAT_30_polar emerges as the highest-ranked feature overall, followed by WT1 163542, WAT_24_polar, WAT_42, and WAT_45. Manual inspection confirms that WAT_30_polar is consistently populated with no missingness, indicating that the aggregated ranking has surfaced a strong and reliable signal.

The dominance of WAT parameters in the aggregated ranking shows that wafer acceptance tests provide stronger and more consistent predictive information than many individual die-level tests (Wang et al.). At the same time, cross-model convergence on WAT 42 and WAT 45 identifies a small set of universally important tests that should receive priority in engineering review and ongoing monitoring (Kang et al.; Lenhard et al.).

Decision Tree Root Cause Analysis Model

The decision tree RCA model converts the aggregated top-fifty feature rankings into clear, interpretable rules that highlight high-risk regions in test space and support adaptive screening and process control (Lenhard et al.; Li et al.; Bergeret and Le Gall). Using the same stratified train–test splits as earlier models, the tree is tuned with RandomizedSearchCV to balance interpretability and performance, with no feature standardization applied so that split thresholds remain meaningful. The final model (Figure 14) relies primarily on WAT 24, WT1 1, WAT 44, WAT 35, and WAT 42 to differentiate success from failure where orange branches indicate failure-leaning regions and blue branches indicate success.

The tree achieves 91% precision and 94% recall for successes and 33% precision with 25% recall for failures, reflecting the inherent difficulty of predicting rare failure events. Its main value lies in its transparency: it reveals the exact cutoff values and parameter combinations most

associated with Category C risk. Scatter plots with decision boundaries (Figure 15) further illustrate how thresholds, such as low WAT_30_polar values, correspond to sharp increases in failures, providing practical guidance for tightening specifications, prioritizing wafer inspections, and designing adaptive screening strategies near high-risk operating regions. These visual decision boundaries provide direct guidance for tightening specifications, flagging wafers for enhanced inspection, and designing adaptive screening strategies near high-risk regions (Kang et al.; Lakin et al.).

Conclusion

Our research provides an improved methodology of ensemble feature selection plus decision-tree RCA and actionable insights, including a prioritized list of tests, interpretable thresholds, and validated evidence that multivariate modeling uncovers rich information. These findings provide a quantitatively grounded basis for strategic and operational recommendations to enhance yield diagnostics, improve test strategy, and better allocate engineering attention to the electrical tests that matter most for emerging sources of failure in their semiconductor mass production.

Recommendations

Renesas serves automotive and industrial markets that require near-zero defects and long-term reliability. Our results suggest that Category C yield loss is driven by diagnosable, systematic mechanisms, meaning better diagnostics directly reduce scrap, field risk, and customer oversight. Demonstrating multivariate analytics and decision-tree RCA strengthens Renesas's credibility during customer audits and signals manufacturing maturity. The decision tree model is particularly valuable because it converts statistical relationships into concrete threshold-based

rules that define failure-prone regions, enabling early warning systems and targeted process adjustments.

Recommendation 1: Implement Operational Controls for First Wafer Instability

First wafers show elevated failure rate (~28%) relative to subsequent wafers, indicating unstable start-of-lot conditions. Renesas should introduce tool warm-ups, dummy runs, and automated parameter checks before production begins, or sequence low-risk wafers first. Success is measured by reducing first-wafer scrap below 15% and improving lot stability. The trade-off is added cycle time, requiring cost-benefit alignment with manufacturing leadership.

Recommendation 2: Establish Structured Probe Card Monitoring

Probe Cards 02 and 03 show consistently higher failure rates, indicating hardware-driven yield distortion. Renesas should implement probe card health monitoring to keep all cards within ±10% of fleet-average failure rates and reduce false excursions. Challenges include clarifying ownership and funding metrology, but benefits include cleaner data and more accurate attribution of yield loss.

Recommendation 3: Use Decision Tree Thresholds and Top Features to Guide Investigations

Critical predictors such as WAT_30_polar, WAT_24_polar, WAT_42, WAT_45, and WT1 163541/163542 should anchor focused engineering investigations. Decision tree thresholds indicate where failure risk is highest and can be translated into specification limits and early-warning bands. High-risk nodes should trigger review of equipment logs, recipes, and materials. Success is measured by identifying actionable process adjustments and deploying threshold-based monitors. The primary constraint is engineering time.

Recommendation 4: Expand Models to Include Spatial and Process Variables

Edge patterns, first-wafer effects, and lot variation suggest spatial and equipment conditions are also key drivers. Renesas should incorporate die coordinates, zone identifiers, equipment IDs, chamber states, and recipe versions into next-generation models. This can reveal interactions between electrical, spatial, and process variables, and pinpoint tools, chambers, or regions contributing to failures. The main challenges are integrating equipment and spatial data and validating more complex models.

Recommendation 5: Build Integrated Data Infrastructure and Cross-Functional RCA Teams

Scaling multivariate RCA requires an integrated yield data warehouse linking test data, wafer maps, probe card logs, and equipment information. Cross-functional RCA teams that unite process, test, data, and quality engineers should maintain shared metrics, track validated thresholds, and drive adoption of decision-tree methods. Success indicators include faster root-cause resolution, improved productivity, and expanded libraries of validated thresholds. Barriers to implementation include budget, silos, and training needs; a focused pilot can demonstrate ROI.

Implementation Roadmap

Renesas should initially focus on immediate fixes (first wafer controls, probe card monitoring) and initiating investigations into top features such as WAT_30_polar, WAT 42 and WAT 45. The organization can scale proven changes and build infrastructure for automated wafer map analysis, threshold-based scoring, and integrated pipelines of spatial, equipment, and test features (Jeong et al.; Langford et al.; Wu et al.). Longer-term, the goal is full deployment of multivariate

RCA, with decision-tree threshold libraries, integrated datasets, and mature RCA teams. This transformation from univariate to multivariate RCA, and specifically the adoption of decision tree threshold-based analysis, positions Renesas to meet increasingly strict automotive quality requirements, accelerate product ramps, and reduce manufacturing costs through targeted scrap reduction (Nakata et al.; Lenhard et al.). For customers, improved outgoing quality and transparent analytics capability reduce perceived supply risk. For competitors, Renesas develops a capability advantage in yield management that compounds as validated thresholds and successful interventions accumulate. These implementations represent a strategic investment in manufacturing intelligence that can benefit current and future product generations at Renesas.

Appendix A: Model Hyperparameter Specifications

Lasso, Random-Forest and Decision Tree algorithms are implemented using scikit-learn, with the XGBoost package used for gradient boosting. A consistent preprocessing pipeline includes median imputation (SimpleImputer) for missing values and feature standardization (StandardScaler), except for the decision tree RCA model, which intentionally skips scaling to preserve interpretability of thresholds. To address class imbalance (approximately 15 percent failures), all models use class weights in a 1:10 ratio favoring the failure class. Hyperparameters are optimized with RandomizedSearchCV using ten random iterations, five-fold stratified cross-validation, and weighted F1-score as the scoring metric.

Lasso Logistic Regression

- Regularization parameter C: searched over [1, 10, 100, 1000].
- Optimal C: 10 (moderate L1 penalty strength).
- Solver: saga, penalty: L1.
- Class weight: {0: 1, 1: 10}.
- Maximum iterations: 5000.

Random Forest Classifier

- Number of trees: searched over [200, 400, 600, 800].
- Maximum depth: searched over [4, 6, 8].
- Optimal configuration: 200 trees, maximum depth 8.
- Class weight: {0: 1, 1: 10}.

XGBoost Classifier

- Number of estimators: searched over [200, 400, 600, 800].
- Maximum tree depth: searched over [4, 6, 8].
- Learning rate: searched over [0.001, 0.01, 0.1, 1.0].
- Class imbalance adjustment: scale_pos_weight = 10.
- Optimal configuration: 600 estimators, maximum depth 4, learning rate 0.1.

Decision Tree for RCA

- Maximum depth: searched over [4, 6, 8].
- Minimum samples split: searched over [2, 4, 6, 8].
- Criterion: Gini impurity.
- Class weight: {0: 1, 1: 10}.
- Scaling: intentionally skipped to preserve interpretability of split thresholds.

All models use SimpleImputer with median strategy for missing values and are trained only on features with less than 10 percent missingness. The top fifty features based on aggregated importance scores serve as inputs to the final decision tree RCA model.

Appendix B: Figure Captions

Figure 1. Overall failure distribution by category. Category C dominates probe-stage failures and accounts for about fifteen percent of total die failures.

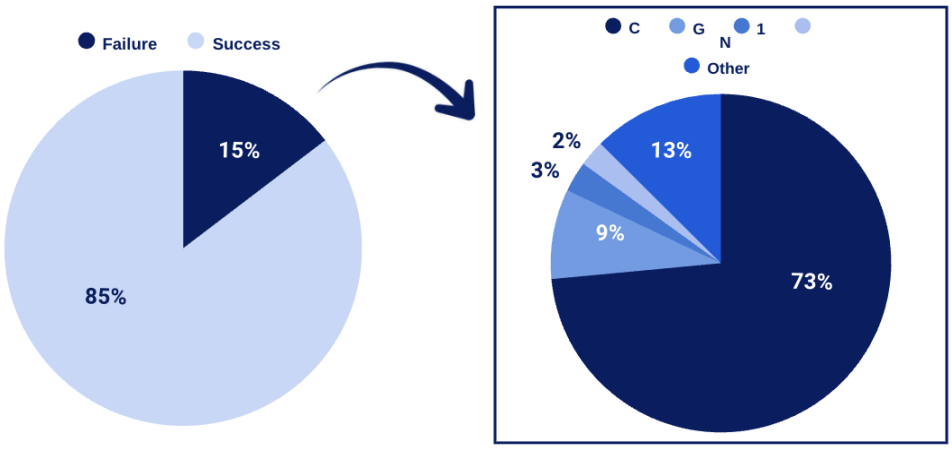


Figure 2. Failure rate by wafer position within each lot. First-wafer failures are consistently higher (about 28%), indicating start-up instability.

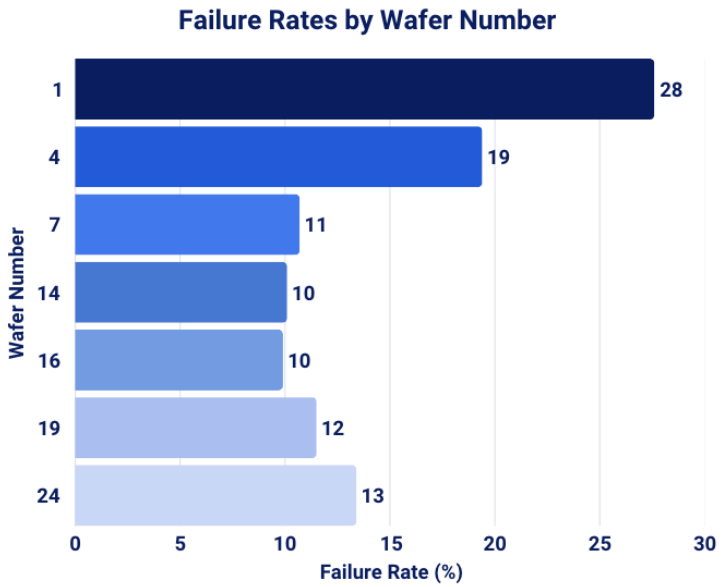


Figure 3. Wafer map with Zone 4 (edge) defect concentration. Failures concentrate at the wafer edge, consistent with edge-related process non-uniformity.

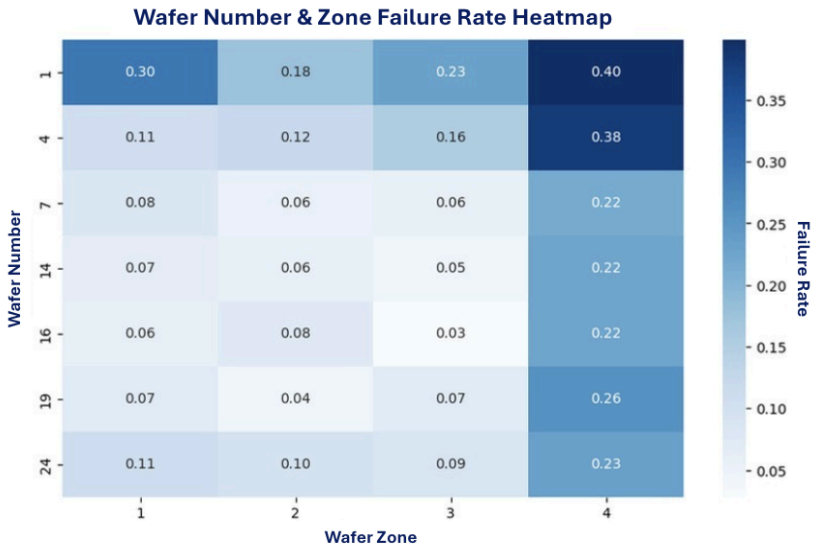


Figure 4. Failure rate by probe card ID. Probe Cards 02 and 03 exhibit elevated failure rates, suggesting probe card degradation.

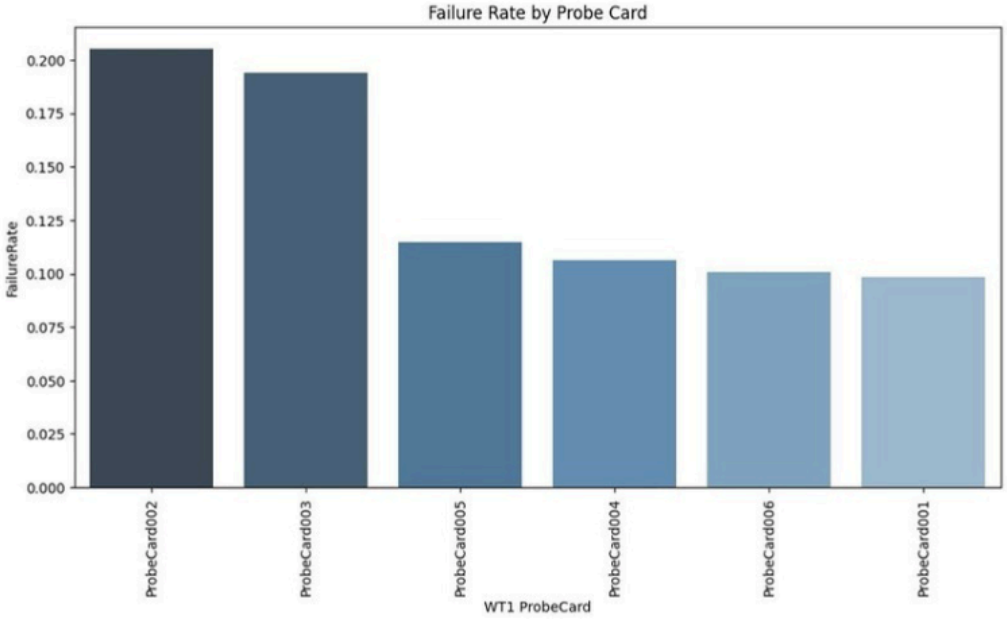


Figure 5. Monthly failure rate trends. A January spike in Category C failures, linked to WT1 test 154001, indicates seasonal or maintenance-related sensitivities.

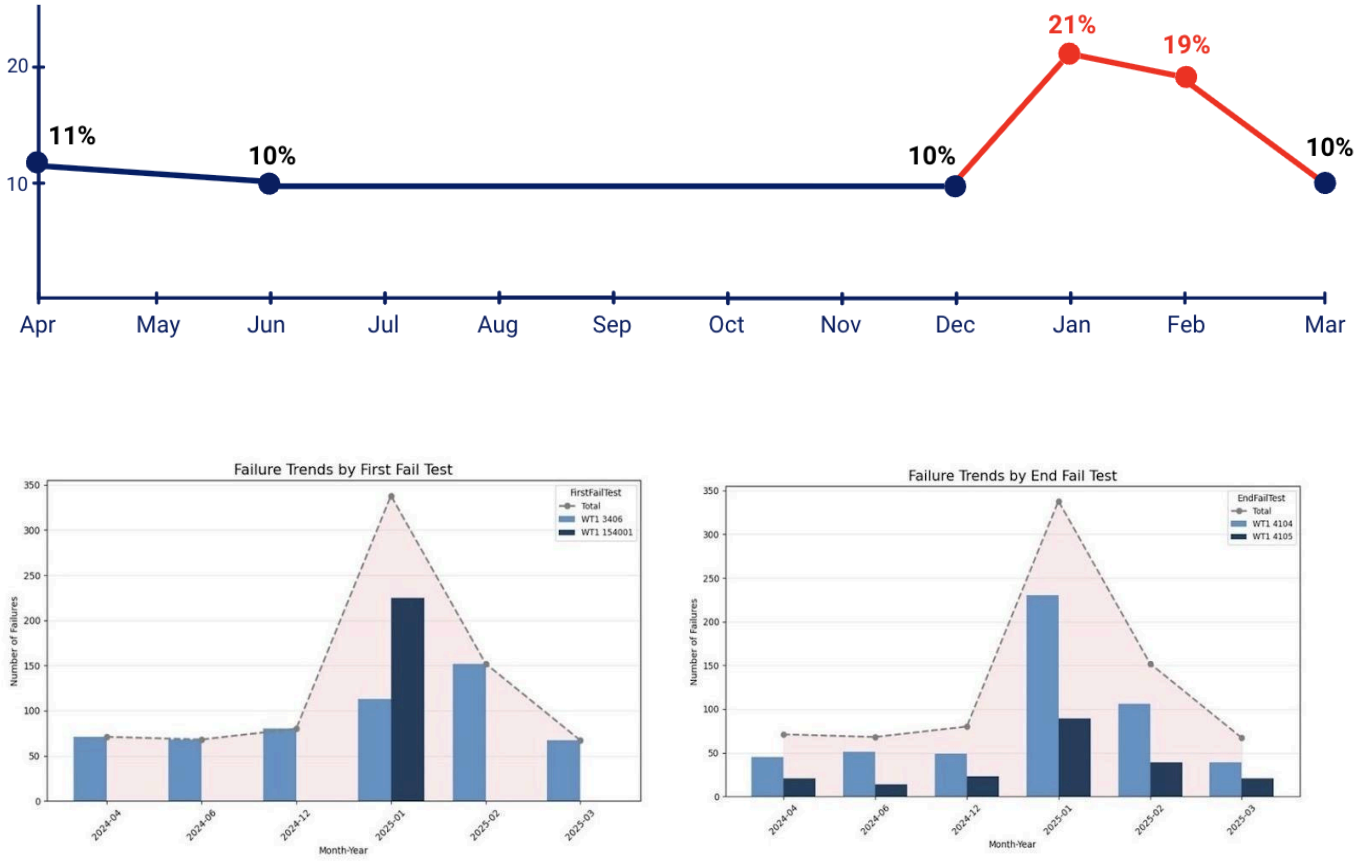


Figure 6. Lot-level failure rate distribution. Lots 2 and 4 show significantly elevated failure rates, pointing to batch- or tool-level instability.

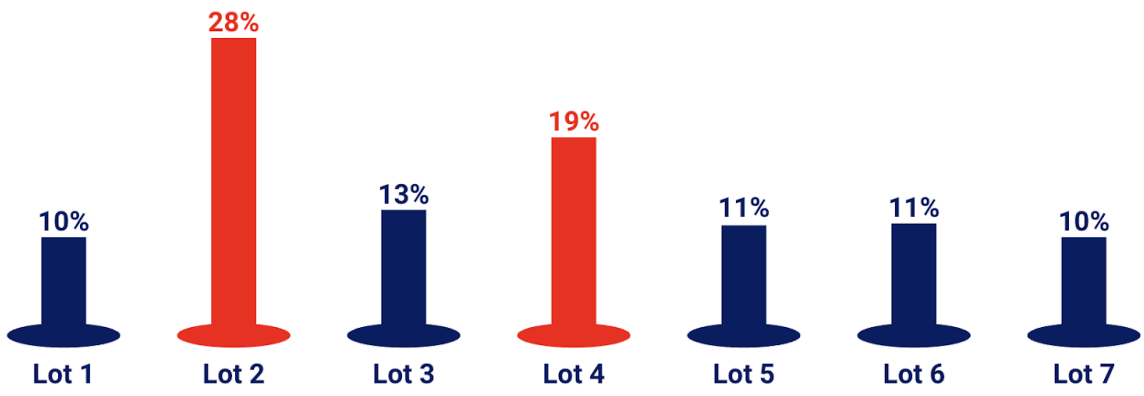


Figure 7. Missingness by Die Outcome Plot

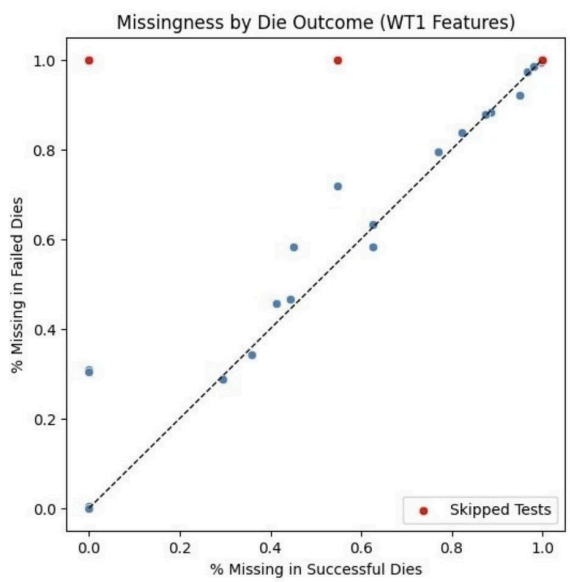


Figure 8. Random Forest feature importance rankings. WAT_30_polar, WAT_14_polar, and WAT_24_polar dominate the ranking.

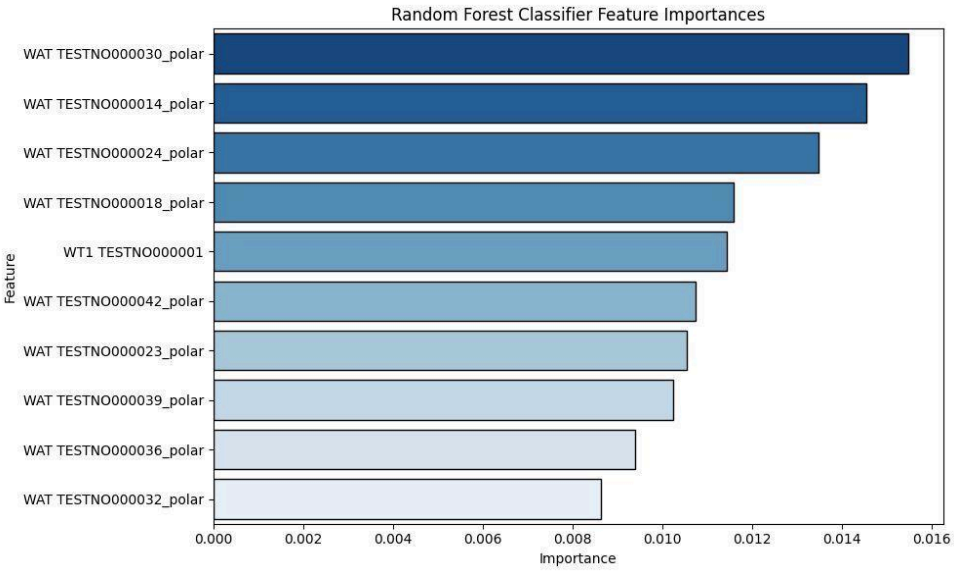


Figure 9. XGBoost feature importance rankings. WAT_30_polar is independently confirmed as the dominant predictor.

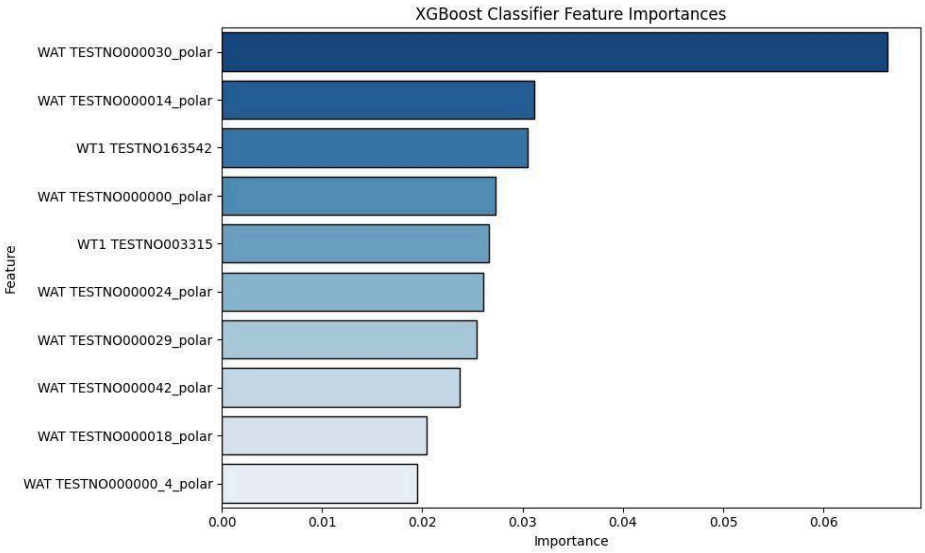


Figure 10. Model hyperparameter tuning and F1-score comparison. Random Forest achieves the highest weighted F1-score, followed by XGBoost and Lasso.

Model	Optimal Parameters	F1-Score
Lasso	L1 Penalty: 0.1	0.34
Random Forest	# of Trees: 200 Max Depth: 8	0.5
XGBoost	# of Trees: 600 Max Depth: 4 Learning Rate: 0.1	0.45

Figure 11. Feature selection overlaps across models. WAT 42 and WAT 45 appear in all three models' top fifty feature lists.

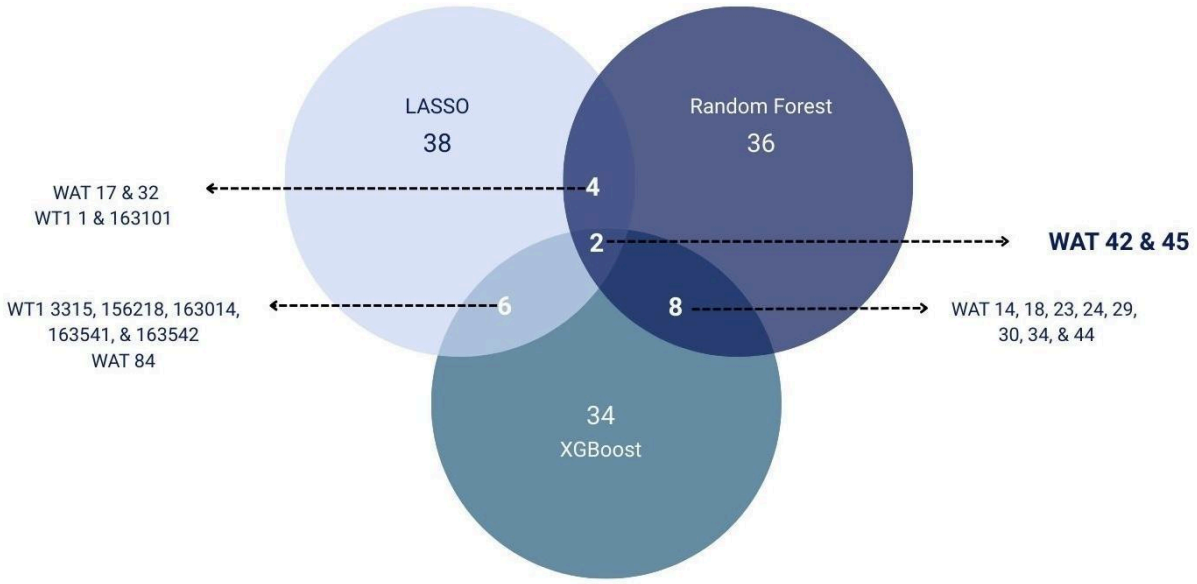


Figure 12. ROC curve comparison across models. All models achieve similar AUC values (0.76 to 0.78), confirming robust predictive signals.

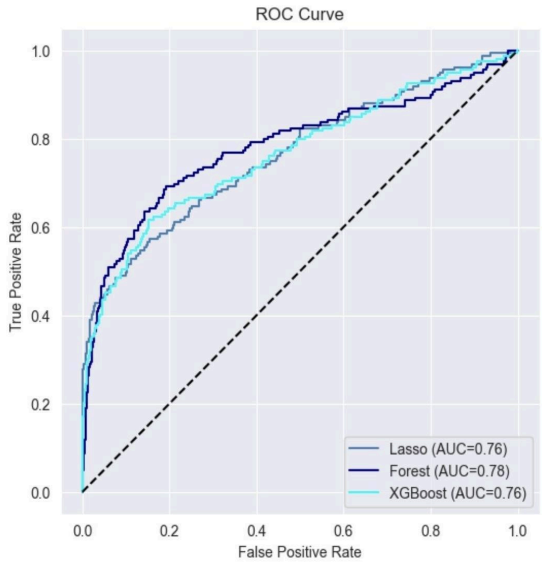


Figure 13. Aggregated feature importance rankings. WAT_30_polar and other WAT parameters dominate the aggregated ranking.

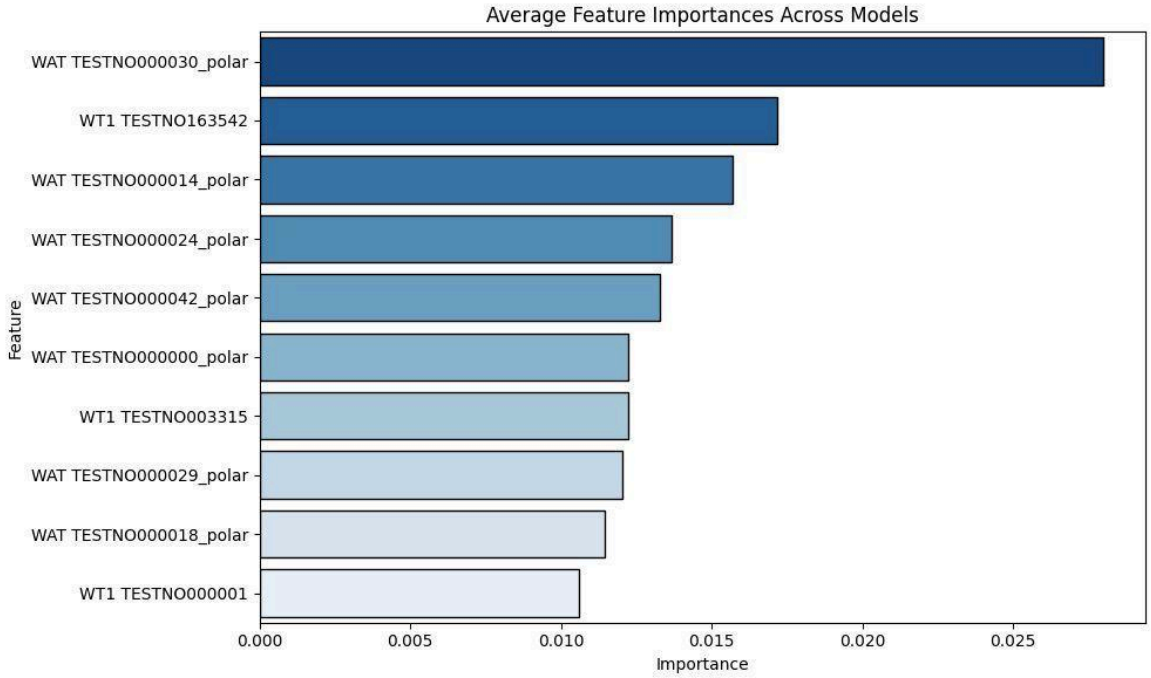


Figure 14. Decision tree RCA model using top fifty features. A small set of parameters and thresholds defines the most failure-prone regions.

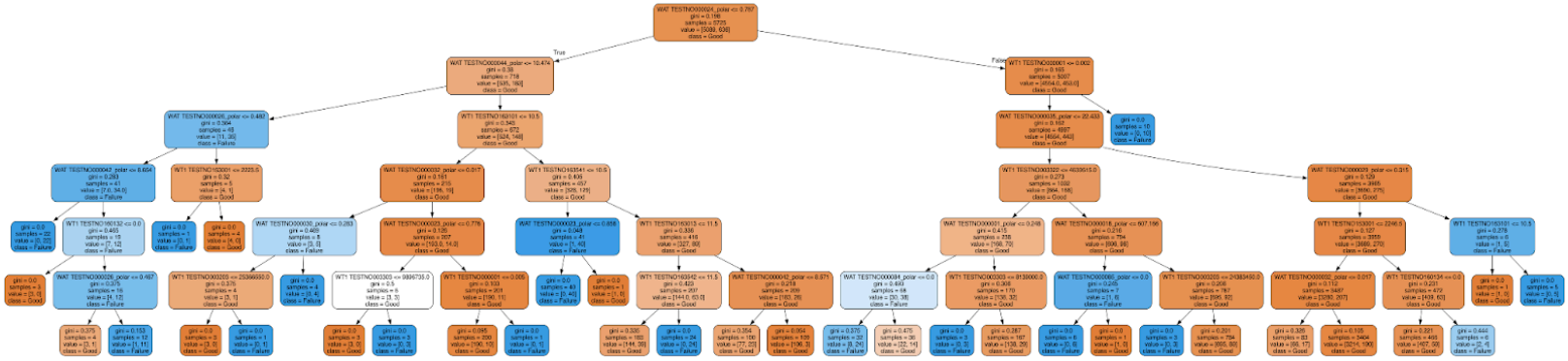
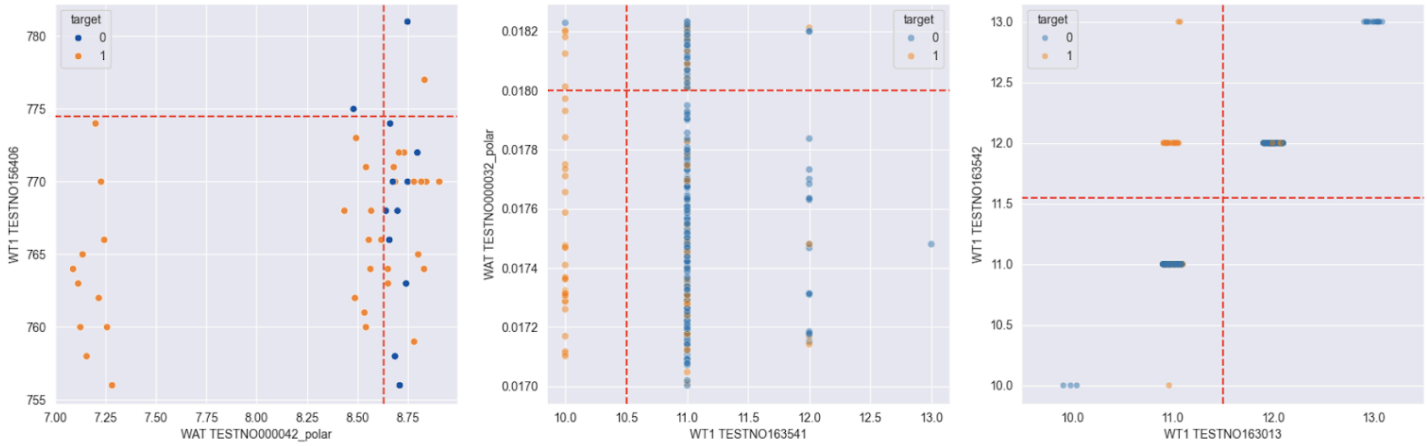


Figure 15. Scatter plots of key feature pairs with decision thresholds. Visual decision boundaries separate passing and failing dies, enabling targeted screening.



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